



04-08-2026

FIRST SEMESTER 2026-27
COURSE HANDOUT

Course Number: EEE G513

Course Title: Machine Learning for Electronics Engineers

Instructor-in-charge: Dr. Ritesh Vyas

Scope of the course:

The rise of machine learning (ML) has introduced many opportunities in the design of electronic systems. This course provides a requisite conceptual foundation of ML and contextual application of ML techniques for improving the performance of electronic systems. The course introduces the major approaches to ML such as supervised, unsupervised, semi-supervised, and reinforcement learning. The course also lays emphasis on various deep learning techniques and their application to various types of electronic systems/sub-systems (e.g. control-dominated systems, NLP systems, vision-based systems, communication systems, embedded systems, IoTs). Additionally, the course covers advance topics in ML such as multi-modal, multi-task learning as well as transfer learning. The course discusses the challenges in implementation of ML techniques, complexity analysis of the ML architectures for hardware implementation, efficient architectures, and various topologies used for implementation.

Outcomes:

- Students should be able to apply ML techniques to model, design, validate, and optimize embedded electronic systems in their area of interest.
- Develop the skills and practical exposure in modeling, developing, and debugging ML-based systems.
- Develop the foundation required to explore advanced topics of research in ML.

Text Books:

T1. I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT Press, 2016.

T2. C. M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006.

Reference Books:

R1. Tom M. Mitchell, "Machine Learning", McGraw-Hill Education (INDIAN EDITION), 2017.

R2. Amanda Casari, Alice Zheng, "Feature Engineering for Machine Learning", O'Reilly, 2018.

R3. Andreas Muller, "Introduction to Machine Learning with Python: A Guide for Data Scientists", Shroff/O'Reilly; 2016.

R4. Shai Shalev-Shwartz and Shai Ben-David, "Understanding Machine Learning: From Theory to Algorithms", Cambridge University Press, 2014.





R5. Aurelien Geron, “Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow”, O’Reilly, 2019.

R6. Simon Haykin, “Neural Networks and Learning Machines”, Third Edition, Paper Back, Pearson India

R7. Richard Sutton, Andrew Barto, “Reinforcement Learning: An Introduction”, The MIT Press, Second edition, 2018.

Course Structure:

1. Introduction to Machine Learning

- 1.1. ML Categories (Supervised / Unsupervised / Semi-Supervised and Active / Transfer and Multitask / Reinforcement / Bayesian / Instance-Based Learning)
- 1.2. Supervised Learning (Techniques other than Deep Learning)
 - 1.2.1. Regression: Linear Regression, Polynomial Regression
 - 1.2.2. Classification: PCA, Probabilistic PCA, Decision Tree, Support Vector Machines, Probabilistic Generative Classifier (Bayes), and Discriminative Classifiers (Logistic Regression)
- 1.3. Unsupervised Learning
 - 1.3.1. K-means Clustering, EM Clustering
- 1.4. Data Types, Data Tagging, Labelling
- 1.5. Feature Selection, Transformation, Dimensionality Reduction
- 1.6. Hardware Platforms, Tools, and Software Packages for ML (with emphasis on tools and packages required for lab exercises)

2. Basics of Neural Networks

- 2.1. Multilayer Perceptron (MLP)
 - 2.1.1. Basics: Universal Approximation, Theorem, Activation functions
 - 2.1.2. Supervised Training (Input Space, Samples, Normalization/Denormalization, Forward Propagation, Loss Functions, Error Computation, Backpropagation, Parameter Update, Generalization)
 - 2.1.3. Model Training (Early stopping, Regularization, Hyperparameter Selection, Evaluation Metrics Model Performance,)

3. Convolutional Neural Networks

- 3.1. Introduction and Building Blocks of CNN
 - 3.1.1. Convolution and Co-relation, Convolution Types
 - 3.1.2. Activation Functions-Types
 - 3.1.3. Pooling Layer Types and Subsampling
 - 3.1.4. Local Connectivity or Sparse Interactions, Parameter Sharing
 - 3.1.5. Spatial Arrangement
 - 3.1.6. Classical CNN Architecture
- 3.2. CNN Learning and Training
 - 3.2.1. Weight Initialization
 - 3.2.2. Regularization of CNN
 - 3.2.3. Gradient Computation and Gradient-based Learning





3.2.4. Optimizers

3.2.5. Reusing Pretrained Layers

3.3. Other CNN Architectures

4. Recurrent Neural Networks

4.1. Sequence Data and Modelling

4.1.1. Discrete-time Markov Chains

4.1.2. Hidden Markov Model (Discriminative Approach)

4.1.3. Conditional Random Fields (Generative Approach)

4.2. Introduction, Building blocks of RNN

4.2.1. Recurrence and Memory

4.2.2. Forward and Backward Propagation in RNNs

4.2.3. Vanishing Gradient Problem & Regularization

4.3. RNN Architectures*

4.3.1. Classical Deep RNN

4.3.2. Bidirectional RNN

4.3.3. Residual LSTM

4.3.4. Recurrent Highway Networks

4.3.5. SRU and Quasi RNN

4.3.6. Recursive Neural Network

*1/2 will be covered in detail and the introduction to the rest of the types may be provided or assigned as self-study.

5. Generative Deep Learning*

5.1. Energy-Based Models (Introduction)

5.2. Boltzmann & Restricted Boltzmann Models

5.3. Deep Belief Networks

5.4. Autoencoders, Variational Encoder

5.5. Sparse Coding (Introduction)

5.6. Generative Adversarial Networks (GANs)

*1/2 will be covered in detail and the introduction to the rest of the types may be provided or assigned as self-study.

6. ML Hardware Architectures

6.1. Choosing hardware architecture for Machine/Deep Learning: FPGA, GPU, Microcontrollers, MPSoC; Embedded System Considerations

6.2. Complexity Analysis of DNN, Performance Tradeoffs: Memory, Latency, Throughput, Power Consumption

6.3. Efficient Topologies, Quantization and Accuracy Tradeoffs, Pruning (Memory and Energy-aware)

7. Advanced Topics

7.1. Transfer Learning: Introduction, Applications

7.2. Deep Reinforcement Learning: Introduction, Applications

7.3. Multi-Modal and Multi-Task Learning

7.3.1. Handling Multiple Modalities e.g. Text and Image, Text and Speech, Audio and Video

7.3.2. Multi-Task Learning e.g. Speech, NLP, or Image Domains





7.4. Current Status and Future Trends

Lecture Plan:

Lecture No.	Topic	Learning Objective	Text/Ref. Book
1-2	Introduction to ML	Gain foundational understanding of machine learning concepts	T1, R1
3-15	Categories of ML	Gain insights of supervised (regression, classification, PCA, SVM), unsupervised (k-means, GMM, EM), and reinforcement (MDP, DP, MC, TD) learning	T1, T2, R1, R2, R7
16-25	Neural Networks and Deep Learning	MLP, BPN, GD, CNN, RNN, Transfer learning, GAN, VAE	T1, T2, R6
26-35	Use Cases & Applications	Diverse applications based on CNN, RNN	R3, R4, R5, published research papers
36-41	ML Hardware	complexity analysis of the ML architectures for hardware implementation, efficient architectures topologies for ML implementation	R3, R4, R5, Published research papers
42	Concluding Lecture		



Evaluation Scheme:

Component	Weightage	Date & Time	Evaluation type
Mid Sem	25%	As decided by AGSRD	Closed Book
Assessment Tasks	20%	TBA	Open Book
Projects	20%	TBA	Open Book
Comprehensive Exam	35%	As decided by AGSRD	Closed Book

Chamber Consultation Hour: Tuesday, 5-6 PM (Chamber: 2219-E)

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Notices: All notices will be displayed on the course website.

Make-up Examination: Make-up examination will be given only in extremely genuine cases for which prior permission of the instructor-in-charge is required.

Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable. It shall be the responsibility of the individual student to be regular in attending lectures as per the schedule announced in the timetable.

Dr. Ritesh Vyas
Instructor-in-charge
Course No. EEE G513